```
> restart:

  VerifyLemma23 := proc()
      local Nmax, Ptab, Atab, Btab,
            n, r, g, sgn, total, t, s, k,
            X0, Y0, X, Y, uv, baseIdx, idx, value,
            labels, minima, argmins, counts,
            expectedMinima, expectedAt, expectedCounts;

      # The largest required index is 2*9999+4 = 20002.
      Nmax := 20002;

      # Exact partition numbers.
      Ptab := Array(0..Nmax, datatype=anything, fill=0);
      Ptab[0] := 1;

      # Euler's pentagonal recurrence.
      for n from 1 to Nmax do
          total := 0;
          r := 1;
          sgn := 1;

          while true do
              g := iquo(r*(3*r-1), 2);

              if g > n then
                  break;
              end if;

              total := total+sgn*Ptab[n-g];

              if g+r <= n then
                  total := total+sgn*Ptab[n-g-r];
              end if;

              r := r+1;
              sgn := -sgn;
          end do;

          Ptab[n] := total;
      end do;

      # a(k) = p(k)-p(k-1), b(k) = p(k)-p(k-2).
      # The convention is p(k)=0 for k<0.
      Atab := Array(0..Nmax, datatype=anything, fill=0);
      Btab := Array(0..Nmax, datatype=anything, fill=0);
```

```
Atab[0] := 1;
Btab[0] := 1;

for k from 1 to Nmax do
    Atab[k] := Ptab[k]-Ptab[k-1];

    if k = 1 then
        Btab[k] := Ptab[k];
    else
        Btab[k] := Ptab[k]-Ptab[k-2];
    end if;
end do;

labels := ["E", "U_R0", "V_R0", "U_Rs", "V_Rs"];

minima := Array(1..5, datatype=anything, fill=0);
argmins := Array(1..5, datatype=anything, fill=0);
counts := Array(1..5, datatype=anything, fill=0);

# Check E(t) > 0 for 72 <= t <= 9999.
for t from 72 to 9999 do
    value := Atab[t]^2-Atab[t-1]*Atab[t+1]
             -2*Atab[t]+Atab[t-1]+1;

    if value <= 0 then
        error "Nonpositive E(t): [t,E(t)]=%1", [t,value];
    end if;

    if argmins[1] = 0 or value < minima[1] then
        minima[1] := value;
        argmins[1] := t;
    end if;

    counts[1] := counts[1]+1;
end do;

X0 := Atab[72];
Y0 := Atab[71];

if X0 <> 695578 or Y0 <> 609237 then
    error "Incorrect reference values: [a(72),a(71)]=%1",
          [X0,Y0];
end if;

# U_s(X,Y) = Y*b(s)-X*b(s-2).
```

```
    # V_s(X,Y) = Y^2*b(s+2)-2*X*Y*b(s)+X^2*b(s-2).
    # For Y>0, these are denominator-cleared forms of:
    # (X/Y)*b(s-2) < b(s), and q_s(X/Y) > 0.
    for s from 3 to 9999 do
        if s <= 34 then
            X := X0;
            Y := Y0;
            baseIdx := 2;
        else
            X := Atab[2*s+4];
            Y := Atab[2*s+3];
            baseIdx := 4;
        end if;

        if Y <= 0 then
            error "Nonpositive ratio denominator at s=%1", s;
        end if;

        uv := [
            Y*Btab[s]-X*Btab[s-2],
            Y^2*Btab[s+2]-2*X*Y*Btab[s]+X^2*Btab[s-2]
        ];

        for k from 1 to 2 do
            idx := baseIdx+k-1;
            value := uv[k];

            if value <= 0 then
                error "Nonpositive certificate:
[name,s,value]=%1",
                      [labels[idx],s,value];
            end if;

            if argmins[idx] = 0 or value < minima[idx] then
                minima[idx] := value;
                argmins[idx] := s;
            end if;

            counts[idx] := counts[idx]+1;
        end do;
    end do;

    # Exact minima and first attaining indices from Appendix A.
    expectedMinima := [
        27986845,
        522896,
```

```
            168215963182,
            774894329,
            149021112963685
        ];

        expectedAt := [73, 3, 4, 35, 35];
        expectedCounts := [9928, 32, 32, 9965, 9965];

        for k from 1 to 5 do
            if minima[k] <> expectedMinima[k]
               or argmins[k] <> expectedAt[k]
               or counts[k] <> expectedCounts[k] then
                error "Certificate mismatch:
    [name,count,minimum,index]=%1",
                      [labels[k],counts[k],minima[k],argmins[k]];
            end if;

            print([
                labels[k], "PASS",
                "checks", counts[k],
                "minimum", minima[k],
                "first_index", argmins[k]
            ]);
        end do;

        print(["LEMMA_2_3_AUXILIARY_CERTIFICATE_PASS"]);
        return true;
    end proc:

    CertificateResult := VerifyLemma23():
```

```
            ["E", "PASS", "checks", 9928, "minimum", 27986845, "first_index", 73]
            ["U_R0", "PASS", "checks", 32, "minimum", 522896, "first_index", 3]
          ["V_R0", "PASS", "checks", 32, "minimum", 168215963182, "first_index", 4]
          ["U_Rs", "PASS", "checks", 9965, "minimum", 774894329, "first_index", 35]
       ["V_Rs", "PASS", "checks", 9965, "minimum", 149021112963685, "first_index", 35]
                    ["LEMMA_2_3_AUXILIARY_CERTIFICATE_PASS"]
[ >
```